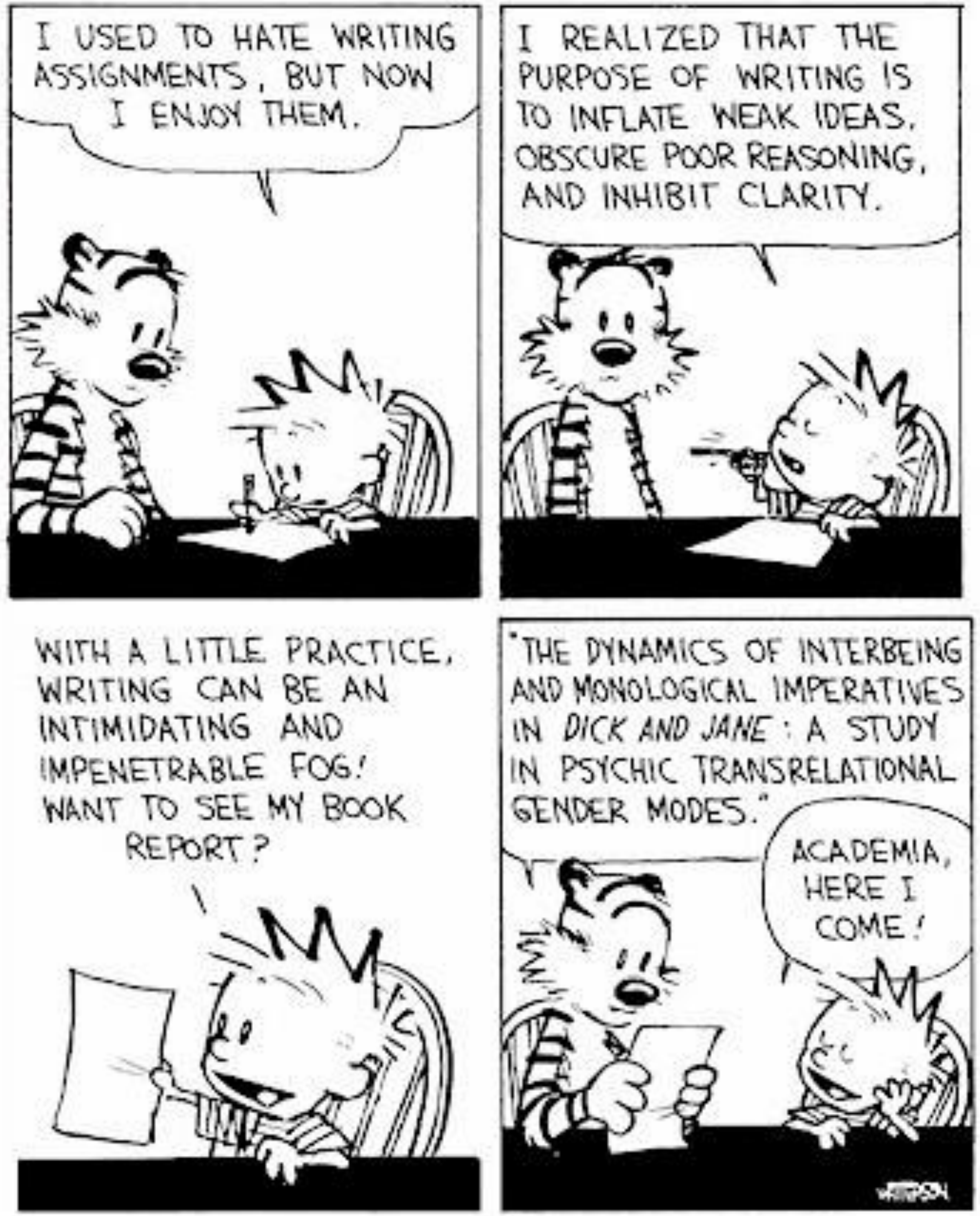
# How to write a great research paper

# Why bother?

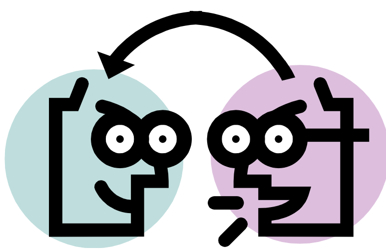
## Fallacy

we write papers and give talks mainly to impress others, gain recognition, and get promoted




# Papers communicate ideas

- Your goal: to infect the mind of your reader with **your idea**, like a virus
- Papers are far more durable than programs



The greatest ideas are (literally) worthless if you keep them to yourself

# Writing papers: model 1

Idea → Do research → Write paper

- ❑ Natural approach

# Writing papers: model 2

Idea → Write paper → Do research

- Forces us to be clear, focused
- Crystallises what we don’t understand
- Opens the way to dialogue with others: reality check, critique, and collaboration

# Do not be intimidated

- **Writing the paper is how you develop the idea in the first place**
- It usually turns out to be more interesting and challenging that it seemed at first

# The purpose of your paper

# The purpose of your paper is...

**To convey your idea**

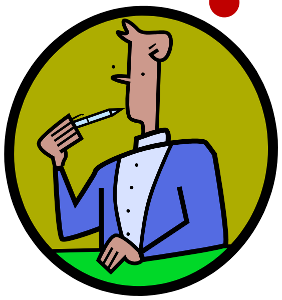
- ... from your head to your reader's head
- Everything serves this single goal

# Conveying the idea

- ❑Here is a problem
- ❑It’s an interesting problem
- ❑It’s an unsolved problem
- ❑**Here is my idea**
- ❑My idea works (details, data)
- ❑Here’s how my idea compares to other people’s approaches

I wish I knew how to solve that!

I see how that works. Ingenious!



# Conveying the idea

1. Many papers are badly written and hard to understand
2. This is a pity, because their good ideas may go unappreciated
3. Following simple guidelines can dramatically improve the quality of your papers
4. Your work will be used more, and the feedback you get from others will in turn improve your research

# Structure

Title (1 paragraph)
Abstract (4 sentences)
Introduction (1 page)
Related work (1-2 pages)
The problem (1 page)
My idea (2 pages)
Results and Discussion (2-5 pages)
Conclusions and future work (0.5 pages)
References

# Title

- Clear and Short (if possible appealing and selective).
- Examples:

Deep Learning for Eye Blink Detection Implemented at the Edge

Robot Path Planning via Neural-Network-Driven Prediction

Flexible Communication System for 6G Based on Orthogonal Chirp Division Multiplexing

An Analysis of Adversarial Attacks and Defenses on Autonomous Driving Models

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# The abstract

- Summary of the paper, i.e., review of the goals or research problems, which research methodology was applied, results achieved and major contribution (100-500 words).
- Often written last
- Used by program committee members to decide which papers to read
- Must be short, for example only four sentences can be enough
    1. State the problem
    2. Say how you address the problem
    3. Say how the solution works
    4. Say what it achieves/what follows from your solution
- Keywords: allow positioning of the paper

# The abstract - Example

Problem

*Abstract*—In autonomous vehicles, accurate extrinsic calibration for LiDAR and camera is an essential prerequisite for multi-sensor information fusion. Automatic and targetless extrinsic calibration has become the mainstream of academic research in recent years. However, existing automatic calibration methods that rely on edge or semantic features are unrobust, or require

Proposed approach

specific scene settings. In this paper, instance segmentation is used for automatic extrinsic calibration of the LiDAR and camera

How it works

for the first time. Key targets from the segmented instances are extracted and correlated. Regarding the extrinsic calibration as an optimization problem, a novel cost function based on the matching degree of the appearance and centroids from the key targets of the point cloud and image pairs is formulated. Subsequently, differential evolution is used to minimize the cost function to obtain the optimal extrinsic parameters.

What it achieves

Extensive experiments on the KITTI dataset and Waymo Open Dataset demonstrate the accuracy and robustness of the proposed method. The MAE of rotation and translation is less than $0.3^\circ$ and 0.05 m respectively, which outperforms semantic-based and edge-based approaches in terms of accuracy.

keywords

*Index Terms*—Sensor Fusion, AI-Based Methods, Computer Vision for Automation

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# The introduction (1 page)

1. Start by very briefly introducing the context/topic
2. Motivate/describe the problem
3. State your contributions

...and that is all

# Context and problem motivation

I. INTRODUCTION

PATH planning, as a fundamental problem in robotics [1], aims at finding a collision-free path that directs the robot to move from the start position to the goal position. It has been applied to many real-world applications, including but not limited to autonomous driving [2], humanoid robots [3], industrial manufacturing, and medical surgery. Many methods have been proposed to address the path planning problem, which can be generally classified into three categories. The grid-based methods discretize the whole state space into a number of grids and then implement the graph search algorithm like A* [4] to find a feasible path. However, they do not perform well in large-scale problems, and different parameter settings have a big impact on the planning result. The artificial potential field (APF) methods [5] use the attractive force from the goal position and the repulsive force from the obstacles to construct a potential field and then direct the robot to follow the steepest descent of the potential. But they suffer from the local minimum problem. The sampling-based methods such as probabilistic roadmap [6] and rapidly exploring random tree (RRT) [7] incrementally sample states from the state space to construct an implicit representation like a graph or tree, with which a feasible path can be found by connecting the start and goal node through edges. However, the generated path is nonoptimal, and the convergence speed to the optimal solution is slow.

# State your contributions

- Write the list of contributions
- The list of contributions drives the entire paper: the paper substantiates the claims you have made
- Reader thinks “gosh, if they can really deliver this, that’s be exciting; I’d better read on”

# State your contributions

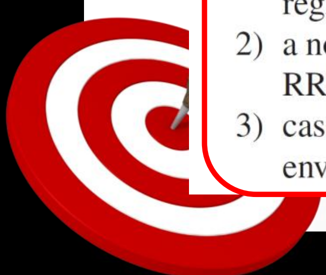
To solve the aforementioned limits of the heuristic methods, in this article, a novel heuristic method named *neural-network-driven prediction (NEED)* is presented. NEED has an encoder–decoder structure, which consists of three modules: convolutional neural network (CNN) backbone, spatial pooling module, and the decoder module. A lot of successful path planning cases are used to train NEED to equip it with the ability of predicting a promising region for the new path planning problem. This predicted region will be used to bias the search direction in the path planning process. Like other heuristic methods, NEED can be applied to the existing path planning algorithm. As shown in Fig. 1, NEED acquires the environment information to make a prior prediction, which is transferred to the path planning algorithm as a heuristic. Meanwhile, the planning algorithm combines the environment information and the generated heuristic to quickly generate a feasible path. It is noted that NEED is not designed for a certain environment or a certain planning algorithm; it can be used in various environments and algorithms.

The contributions of this article are summarized as follows:

1) an efficient neural network to predict the promising search region for path planning algorithms;
2) a novel nonuniform sampling-based method based on the RRT* [9] and NEED;
3) case studies of NEED in both 2-D static and dynamic environments.

Do not leave the reader to guess what your contributions are!

Bulleted list of contributions



# Contributions should be refutable

| NO! | YES! |
|---|---|
| We describe the WizWoz system. It is really cool. | We give the syntax and semantics of a language that supports concurrent processes (Section 3). Its innovative features are... |
| We study its properties | We prove that the type system is sound, and that type checking is decidable (Section 4) |
| We have used WizWoz in practice | We have built a GUI toolkit in WizWoz, and used it to implement a text editor (Section 5). The result is half the length of the Java version. |

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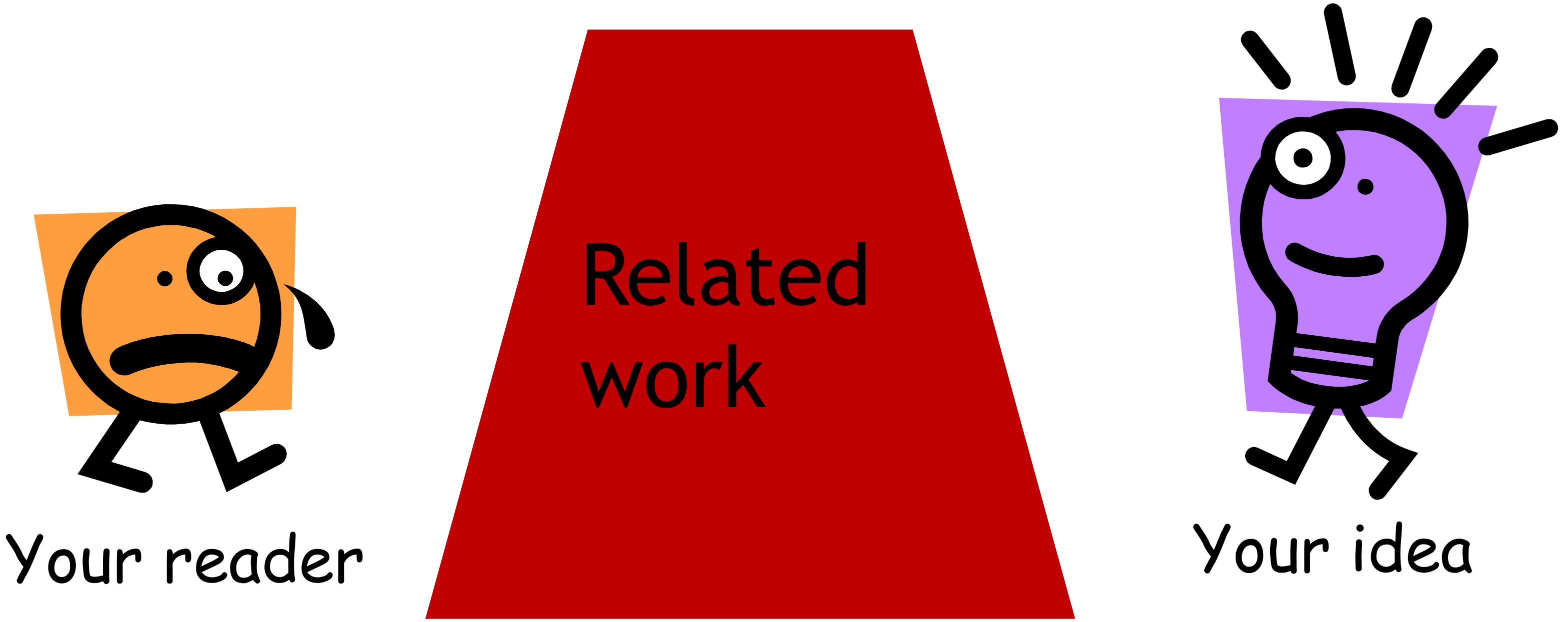
# Related work…




We adopt the notion of transaction from Brown [1], as modified for distributed systems by White [2], using the four-phase interpolation algorithm of Green [3]. Our work differs from White in our advanced revocation protocol, which deals with the case of priority inversion as described by Yellow [4].

# Related work…

- Concentrate single-mindedly on a narrative that
  - Describes the problem, and why it is interesting
  - Describes your idea
  - Defends your idea, showing how it solves the problem, and filling out the details
- On the way, cite relevant work in passing, but defer discussion to the end
- Important to make sure related work is accurate!

# Related work

Fallacy — To make my work look good, I have to make other people's work look bad

# The truth: credit is not like money

**Giving credit to others does not diminish the credit you get from your paper**

- Warmly acknowledge people who have helped you
- Be generous to the competition.

  "In his inspiring paper [Foo98] Foogle shows.... We develop his foundation in the following ways..."
- Acknowledge weaknesses in your approach

Dissertação em Engenharia

# Credit is not like money

Failing to give credit to others can kill your paper

- If you imply that an idea is yours, and the referee knows it is not, then either
  - You don't know that it's an old idea (bad)
  - You do know, but are pretending it's yours (very bad)

# Structure

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# The payload of your paper

Consider a bifurcated semi-lattice D, over a hyper-modulated signature S. Suppose $p_i$ is an element of D. Then we know for every such $p_i$ there is an epi-modulus j, such that $p_j < p_i$.

- Sounds impressive...but sends readers to sleep
- In a paper you MUST provide the details, but FIRST convey the idea
  - It can be helpful to provide EXAMPLES and only then present the general case

 

# Using examples

*2) Antenna indexing:* The selection of $n_{rf}$ out of $n_t$ antennas for transmission is made based on antenna index bits. The antenna index bits choose an 'antenna activation pattern', which tells which $n_{rf}$ antennas out of $n_t$ antennas are used for transmission. There are $l_a = \binom{n_t}{n_{rf}}$ antenna activation patterns possible, and $k_a = \left\lfloor \log_2 \binom{n_t}{n_{rf}} \right\rfloor$ bits are used to choose one among them. These $k_a$ bits are the antenna index bits. Note that not all $l_a$ activation patterns are needed, and any $2^{k_a}$ patterns out of them are adequate. Take any $2^{k_a}$ patterns out of $l_a$ patterns and form a set called the 'antenna activation pattern set', denoted by $\mathbb{S}_a$.

Illustrative Example

*Example:* Let us illustrate this using the following example. Let $n_t = 3$, $n_{rf} = 2$. Then, $l_a = \binom{3}{2} = 3$, $k_a = \left\lfloor \log_2 \binom{3}{2} \right\rfloor = \lfloor \log_2 3 \rfloor = 1$, and $2^{k_a} = 2$. The possible antenna activation patterns are given by $\{[1,1,0]^T, [1,0,1]^T, [0,1,1]^T\}$. The set $\mathbb{S}_a$ is formed by selecting any two patterns out of the above three patterns. For example, $\mathbb{S}_a$ can be $\mathbb{S}_a = \{[1,1,0]^T, [1,0,1]^T\}$. An $n_{rf} \times n_t$ switch connects the RF chains to the transmit antennas. The chosen $n_{rf}$ out of $n_t$ antennas transmit the MIMO-OFDM symbol constructed using the frequency index bits and $M$-ary modulation bits. The active transmit antennas can change from one MIMO-OFDM symbol to the other.

# Conveying the idea

- **Conveying the intuition is primary**, not secondary
- Once your reader has the intuition, he can follow the details (but not vice versa)
- Even if he skips the details, he still takes away something valuable

# Evidence, Results and Discussion

- Your introduction makes claims
- The body of the paper provides **evidence to support each claim**
- Evidence can be: analysis and comparison, theorems, measurements, case studies
- The results achieved should be presented in a clear and resumed format: graphics or tables
- Must include discussion about the results obtained

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# Conclusions and future work

- Should include the answer of the research problem presented earlier.
- It may include conclusions – answers found and summary of the contribution, limitations of the study and directions for future research
- Be brief.

# Conclusions and future work - example

VI. CONCLUSION AND FUTURE WORK

Summary with main Conlusions

In this article, we propose NEED to predict a promising search region for path planning algorithms. Simulation results demonstrate that NEED can significantly improve the algorithm performance when combined with the well-established RRT* algorithm. NEED also shows its ability of predicting the promising region in dynamic environments.

Future work

There are many possible avenues for the future research and applications: first, applying NEED to other existing path planning algorithms to further improve their performance; second, combining psychology and sociology knowledge with NEED can make some human-aware or socially aware path predictions.

# Language and style

# Basic stuff

- Submit by the deadline (Conference, Special Issue, self-defined)
- Keep to the length restrictions
  - Do not narrow the margins
  - Do not **use 6pt font**
  - On occasion, supply supporting evidence (e.g. experimental data, or a written-out proof) in an appendix
- Always use a spell checker (and maybe a translator, ex: https://www.deepl.com/translator)

# Visual structure

- Give strong visual structure to your paper using
  - sections and sub-sections
  - bullets
  - italics
- Find out how to draw pictures/diagrams, and use them

# Visual structure

attack, the AdvGAN model could also be trained on the proxy model. Then universal perturbations and AdvGAN could be integrated into a malware to conduct black-box attacks. Other two attacks are not available under the black-box setting as they need information about the driving model in real-time to construct adversarial examples. Therefore, we only evaluate Universal perturbation attack, AdvGAN attack and AdvGAN universal perturbation attack in black-box attack experiments.

We use a common metric, attack success rate, to measure the performance of adversarial attacks and defenses. An attack is considered successful, if the steer angle deviation is greater than the adversarial threshold $\Delta = 0.3$. An attack success rate is computed as the portion of the number of adversarial examples that the attack is successful out of all generated adversarial examples. All experiments are conducted in a simulation environment with Intel i7-8700 3.2GHz, 32GB of memory, and a NVIDIA RTX 2080Ti GPU.

*A. Research Questions*

We investigate the following research questions:

- **RQ1:** In the white-box setting, how does each of the five attacks perform on different driving models?
- **RQ2:** Do those attacks still perform well in the black-box setting?
- **RQ3:** Do adversarial defense methods improve the robustness of driving models against adversarial attacks?

V. RESULT

This section presents experiment results of five attacks and three defenses on three driving models.

*A. RQ1. Effectiveness of White-box Attacks*

Table II shows the attack success rate of applying five attacks on three selected driving models. All attack methods except IT-FGSM achieve high attack success rate on all three models. The highest attack success rate of IT-FGSM is 59.2% on Epoch model but other methods can achieve over 90% on all three models. Opt_uni and AdvGAN even achieve 100% on Epoch model and average over 99% on the other two models. The reason is that IT-FGSM only perturbs pixels in an image by simply adding the sign of gradients, while Opt and Opt_uni utilize Adam optimizer to search adversarial perturbations in multiple iterations. AdvGAN and AdvGAN_uni learn intrinsic features (e.g., yellow road lanes) that influence steer angle predictions to generate adversarial perturbations. IT-FGSM has the lowest attack success rate on VGG16, implying that the

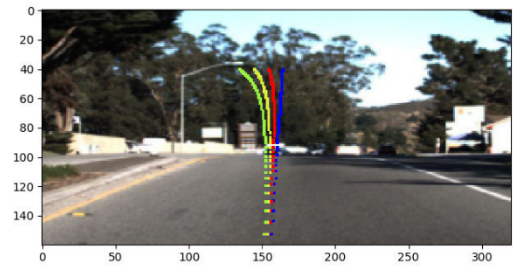


Fig. 3. Display of steering angle track on the driving scene image. The blue, red, yellow and green lines are the tracks with 0, 0.1, 0.2 and 0.3 steering angle deviations respectively. When setting the deviation to 0.3, the vehicle clearly turns to a wrong direction.

TABLE II
ATTACK SUCCESS RATE OF FIVE ATTACKS ON THREE DRIVING MODELS

| | Epoch | Nvidia | VGG16 |
|---|---|---|---|
| IT-FGSM | 59.2% | 31.8% | 16.2% |
| Opt | 91.2% | 99.3% | 95.8% |
| Opt_uni | 100% | 99.9% | 99.9% |
| AdvGAN | 100% | 99.5% | 98.0% |
| AdvGAN_uni | 99.8% | 96.4% | 96.3% |

# Use the active voice

The passive voice is “respectable” but it DEADENS your paper.

| Passive | Active (better) |
|---|---|
| It can be seen that... | We can see that... |
| 34 tests were run | We ran 34 tests |
| These properties were thought desirable | We wanted to retain these properties |

“We” = you and the reader

“We” = the authors

# Use simple, direct language

| NO | YES |
|---|---|
| The object under study was displaced horizontally | The ball moved sideways |
| On an annual basis | Yearly |
| Endeavour to ascertain | Find out |
| It could be considered that the speed of storage reclamation left something to be desired | The garbage collector was really slow |

# The process

- Start early.  Very early.
  - Hastily-written papers get rejected.
  - Papers are like wine: they need time to mature

 

# Summary

If you remember nothing else:

- ❑ **Identify your key idea**
- ❑ **Make your contributions explicit**
- ❑ **Use examples**
- ❑ **Refer related work**

Check examples of publications in your topic! (and in particular in your target Journal/Conference)

Disclaimer:

**ADAPTED FROM: "HOW TO WRITE A GREAT RESEARCH PAPER", SIMON PEYTON JONES, MICROSOFT RESEARCH, CAMBRIDGE**

# Examples

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# Example

IEEE TRANSACTIONS ON ARTIFICIAL INTELLIGENCE, VOL. 3, NO. 3, JUNE 2022 451

## Robot Path Planning via Neural-Network-Driven Prediction

Jiankun Wang, Jianbang Liu, Weinan Chen, Wenzheng Chi, and Max Q.-H. Meng, *Fellow, IEEE*

***Abstract*—This article presents a novel heuristic method named *neural-network-driven prediction (NEED)* for robot path planning problems. Different from classical heuristic methods, NEED is not designed for some specific environments, which can be applied to various environments. NEED has an encoder–decoder structure, which consists of three modules: convolutional neural network backbone, spatial pooling module, and decoder module. By learning from a number of successful path planning cases, NEED can learn to analyze the environment structure and predict the promising search region for the new path planning problem. This predicted region serves as a heuristic to guide the search direction of path planning algorithms. A series of simulation experiments are conducted to demonstrate that NEED significantly improves the algorithm performance on solving new path planning problems. Meanwhile, NEED can also be applied to highly dynamic environments since it can output the prediction results at a speed of over 100 Hz on a laptop.**

***Impact Statement*—In this article, we aim to provide an efficient learning-based method to accelerate the robot path planning process. Classical path planning algorithms first need to perceive the environment and then implement a series of calculation such as collision checking and data storing to generate a feasible path. When facing complex environments, they do not perform well. Our proposed neural network method can predict the promising region, where the feasible path exists for any given environment. This prediction result is used to guide the path planning process so that the algorithm performance can get significantly improved. Apart from sampling-based algorithms, the proposed neural network model can also be extended to other kinds of path planning algorithms.**

***Index Terms*—Artificial intelligence in robotics, intelligent robots, intelligent systems, neural networks, planning.**

### I. Introduction

PATH planning, as a fundamental problem in robotics [1], aims at finding a collision-free path that directs the robot to move from the start position to the goal position. It has been applied to many real-world applications, including but not limited to autonomous driving [2], humanoid robots [3], industrial manufacturing, and medical surgery. Many methods have been proposed to address the path planning problem, which can be generally classified into three categories. The grid-based methods discretize the whole state space into a number of grids and then implement the graph search algorithm like A* [4] to find a feasible path. However, they do not perform well in large-scale problems, and different parameter settings have a big impact on the planning result. The artificial potential field (APF) methods [5] use the attractive force from the goal position and the repulsive force from the obstacles to construct a potential field and then direct the robot to follow the steepest descent of the potential. But they suffer from the local minimum problem. The sampling-based methods such as probabilistic roadmap [6] and rapidly exploring random tree (RRT) [7] incrementally sample states from the state space to construct an implicit representation like a graph or tree, with which a feasible path can be found by connecting the start and goal node through edges. However, the generated path is nonoptimal, and the convergence speed to the optimal solution is slow.

To further improve the algorithm performance, a lot of heuristic methods are presented. They can be applied to either a certain environment or a certain algorithm. For example, in a sparse environment, adding more samples around the line connecting the start and goal position can dramatically reduce the planning time cost for the sampling-based algorithms. But this heuristic cannot be applied to other complex environments. Another example is using the relative distance between the robot and the goal to construct a new repulsive potential function in the APF algorithms [8], so that the problem of goals nonreachable with obstacles nearby can be avoided. Obviously, this heuristic cannot be applied to other path planning algorithms. In fact, the heuristic methods aim at providing an appropriate search direction for the path planning algorithm in the current environment. But this environment-related heuristic is very limited because the users do not know clearly that which heuristic should be used in the current environment, and the proposed heuristic cannot cover all

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Jiankun Wang and Weinan Chen are with the Department of Electronic and Electrical Engineering, Southern University of Science and Technology, Shenzhen 518055, China (e-mail: wangjk@sustech.edu.cn; chenwn@sustech.edu.cn).

Jianbang Liu is with the Department of Electronic Engineering, The Chinese University of Hong Kong, Hong Kong (e-mail: henryliu@link.cuhk.edu.hk).

Wenzheng Chi is with the Robotics and Microsystems Center, School of Mechanical and Electric Engineering, Soochow University, Suzhou 215021, China (e-mail: wzchi@suda.edu.cn).

Max Q.-H. Meng is with the Department of Electronic and Electrical Engineering, Southern University of Science and Technology, Shenzhen 518055, China, and also with Shenzhen Research Institute, The Chinese University of Hong Kong, Shenzhen 518172, China, on leave from the Department of Electronic Engineering, The Chinese University of Hong Kong, 999077, Hong Kong (e-mail: mengqh@sustech.edu.cn).

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# Structure (another …)

- ❑ Title
- ❑ Abstract

1. Introduction
2. Literature Review
3. Method
4. Results
5. Discussion
6. Conclusions

- ❑ References

# Structure (another …)

Microprocessors and Microsystems 74 (2020) 103017

Contents lists available at ScienceDirect

Microprocessors and Microsystems

journal homepage: www.elsevier.com/locate/micpro

## Molecular communication via diffusion with spherical receiver & transmitter and trapezoidal container

Soha Mohamed, Dong Jian*, Liu Hongwei, Zuo Decheng

*School of computer science and technology, Harbin Institute of Technology, Harbin 150001, China*

A R T I C L E  I N F O

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Received 7 November 2019
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*Keywords:*
Molecular communication
Spherical transmitter
Deep neural networks
Support vector machine
Ridge regression
logistic regression
Trapezoidal container

A B S T R A C T

Molecular communication is a new paradigm that works on the principle of the discharge of molecules as carriers of information. The channel of molecular communication is determined by the signal that is received. The designs of the receiver, the modulation, coding, and capacity rely on the models of the received signal. Therefore, it is difficult to model the number of the received modules (until the time $t$) using an analytical approach. Indeed, communication through molecular diffusion is ubiquitous. In particular, the present, closed form of the impulse response of the diffusion channel usually perceives the transmitter as a source point. This research attempts to develop molecular communication via diffusion through a spherical transmitter and receiver and trapezoidal container. The study has been conducted in two parts. These parts include dataset generation using a probability distribution, and the modelling of the received signal – using a multitude of machine learning (ML) techniques. In this study the code is executed using Python language. In addition, the study utilizes deep neural networks, support vector regression, logistic regression, and ridge regression to model the received signal for a spherical transmitter before comparing the performance of all the models. The results of the study have been compared in terms of the models' accuracy to determine the best performing approach. The proposed techniques offer promising results for modelling the received molecules number until time. The proposed technique models the received signal efficiently for longer distances.



### 1. Introduction

Unlike the wireless communication mode, molecular communication is based on the diffusion concept and uses molecules as information carriers between communicating parties. Parameters of concentration, release time, and type of molecule can be employed for signalling by a molecular transmitter. As an outcome, a mechanism must be established for molecule production at the transmitters [11]. According to Wang et al. [28], molecular communication is a bio-inspired technique seeking to enhance traditional communication in nano micro-scale surroundings, and has stimulated a huge concern in both industrial and academic community. With the growth of the bioengineering concept and nanotechnology, the molecular communication-based BANN (body area nano networks) has been initiated and examined to gain practical nano medical applications. Akyildiz et al. [1] stated that to manage complex tasks of medicine in a body area nano network, the nano machines with storage, calculation, communication, and sensing capabilities must interlink with each other through a molecular communication technique. Therefore, any possible damage and incompatibility in the area of body can be reduced essentially to a manageable level through synthetic biology, as well as molecular communication. Felicitti et al. [13] observed that molecular communication is one of the most available techniques of communication to realize body area nano networks for the internet of bio nano things. Indeed, molecular communication seems to be an alternative, leading electromagnetic communication model that enhances communication among nano machines. Singhal et al. [26] observed further that molecular communication has been inspired by biological communication systems and is performed by biological cells.

According to Yilmaz et al. [30], given a molecular communication through a diffusion system, a transmitter node discharges molecules, and the molecules propagate through the surroundings to an extent that they exist at the receiver node. In spherical receiver and transmitter molecules are reflected and obstructed by the transmitter of radius. The molecules are biased towards moving in receiver direction in spherical transmitter due to the transmitter node obstructing body. Every molecules in spherical transmitter is essential to have a greater probability of hitting the receiver. The spherical transmitter and receiver is very essential for molecular communication through. Chou [7] documented that the received molecules constitute a received signal, and the molecule

* Corresponding author.
*E-mail address:* dan@hit.edu.cn (D. Jian).

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